Shivam Kaushik

Researcher @ University of Florida

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RESEARCH EXPERIENCE

Sep 2023 – Present	**Biological Scientist** **Lab Technician** **University of Florida**, Gainesville, USA *Supervisor:* Dr. Diego E. Rincon-Limas	*until Sep 2025*
Jan 2022 – Dec 2022	**Graduate Student** **University of North Carolina**, Chapel Hill, USA *Supervisor:* Dr. Toshi Hige	
Jun 2017 – Dec 2021	**Research Assistant** **Summer Training** **Regional Centre for Biotechnology**, India *Supervisor:* Dr. Pinky Kain	*Aug 2018 - Dec 2021* *Jun 2017 - Aug 2017*

EDUCATION

Jan 2022 – Dec 2022	**Ph.D. Researcher** | Major: ***Neuroscience*** **University of North Carolina**, NC, USA	**[Transcript]**
Jul 2018 – Jul 2020	**Master of Technology** | Major: ***Biotechnology*** GPA: *3.5/4.0* **Maharishi Dayanand University**, India Dissertation: *Mapping novel neural circuits for feeding and foraging in Drosophila melanogaster*	**[Transcript]**
Jul 2014 – May 2018	**Bachelor of Technology** | Major: ***Biotechnology*** GPA: *3.8/4.0* **Maharishi Dayanand University**, India Dissertation: *Dietary modulation of taste behavior & its impact on aging in Drosophila melanogaster*	**[Transcript]**

PUBLICATIONS

Peer-Reviewed Publications

Neurosci Insights 2026	**Shivam Kaushik**, Kartik Singh, Rahul Kumar, Sachin Kumar, Srishti Sanghi, Teiichi Tanimura, Diego E. Rincon-Limas, Pinky Kain, *Activity and State-Dependent Modulation of Salt Taste Behavior via Pharyngeal Neurons in Drosophila Melanogaster*, Neuroscience Insights 2026.
Ann Neurol 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA mir-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, Annals of Neurology.
eNeuro 2023	Andrew M. Davidson, **Shivam Kaushik**, and Toshihide Hige, *Dopamine-Dependent Plasticity Is Heterogeneously Expressed by Presynaptic Calcium Activity across Individual Boutons of the Drosophila Mushroom Body*, eNeuro, 10(10), ENEURO.0275-23.2023.
J Vis Exp 2023	Manal Shakeel, **Shivam Kaushik**, Teiichi Tanimura, Axel Brockmann, and Pinky Kain, *Tracking Sugar-Elicited Local Searching Behavior in Drosophila*, Journal of Visualized Experiments, (201), e65955.
IJMS 2022	Gouri Chopra, **Shivam Kaushik**, and Pinky Kain, *Nutrient Sensing via Gut in Drosophila melanogaster*, International Journal of Molecular Sciences, 23(5), 2694.
Sci Prog 2022	**Shivam Kaushik**, Srishti Sanghi, Sachin Kumar, Rahul Kumar, and Pinky Kain, *Modulation of sugar feeding behavior by Gymnema sylvestre in Drosophila melanogaster*, Science Progress.
Alzheimers Dement 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behavior in Drosophila*, Alzheimer's & Dementia, 17(Suppl 12), e058538.
ACS Omega 2020	Sujeet Thakur, Krishnendu Goswami, Pallavi Rao, **Shivam Kaushik**, Bhanu Singh, Pinky Kain, Shailendra Asthana, Saikat Bhattacharjee, Prasenjit Guchhait, and S. Eswaran, *Fluoresceinated aminohexanol tethered inositol hexakisphosphate: Studies on Arabidopsis thaliana and Drosophila melanogaster and docking with 2P1M Receptor*, ACS Omega, 5(16), 9585–9597.
JEN 2018	**Shivam Kaushik**, Rahul Kumar, and Pinky Kain, *Salt an essential nutrient: Advances in understanding salt taste detection using Drosophila as a Model System*, Journal of Experimental Neuroscience, 12, 1179069518806894.

Publications

BioRixv 2024	**Shivam Kaushik**, Rahul Kumar, Sachin Kumar, Srishti Sanghi, Teiichi Tanimura, Diego E. Rincon-Limas, and Pinky Kain, *Activity and state-dependent modulation of salt taste behavior via pharyngeal neurons in Drosophila melanogaster*, BioRixv 2024.

Book Chapters

IntechOpen 2019	**Shivam Kaushik** and Pinky Kain, *Understanding taste using Drosophila melanogaster*, in *Animal Models in Medicine and Biology* (Eds. Eva Tvrdá and Sarat Chandra Yenisetti), IntechOpen.
IntechOpen 2020	**Shivam Kaushik**, Shivangi Rawat, and Pinky Kain, *A look into the central taste circuits using Drosophila melanogaster as a model system*, in *Role of Obesity in Human Health and Disease* (Eds. Venketeshwer Rao and Leticia Rao), IntechOpen.
IntechOpen 2021	Zoha Sadaqat, **Shivam Kaushik**, and Pinky Kain, *Gut feeding the brain: Drosophila gut an animal model for medicine to understand mechanism mediating food preferences*, in *Preclinical Animal Modeling in Medicine* (Eds. Enkhsaikhan Purevjav, Joseph F. Pierre, and Lu Lu), IntechOpen.

Manuscripts in Preparation

- **Shivam Kaushik** and Diego E. Rincon-Limas, *Optogenetic control of neurodegenerative protein aggregation in Drosophila*.
- Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, and Diego E. Rincon-Limas, *The SARS-CoV-2 NSP1 protein promotes amyloid beta aggregation and toxicity in vivo*.

TALKS & POSTERS

FL Genetics 2025	James Caprio, Riana Shah, Camila Campos, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Investigating the Impact of SARS-CoV-2 Proteins on Neurodegeneration in Fruit Flies*, Florida Genetics Symposium by University of Florida, Gainesville, Florida, USA. November 5–6, 2025.
FL Genetics 2025	Arushi Pant, Jessica Jiang, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Drosophila models of Amyotrophic lateral sclerosis (ALS)*, Florida Genetics Symposium by University of Florida, Gainesville, Florida, USA. November 5–6, 2025.
UF URS 2025	James Caprio, Riana Shah, Camila Campos, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Studying the role of SARS-CoV-2 proteins in neurodegeneration using fruit flies*, Undergraduate Research Symposium organized by University of Florida, Gainesville, Florida, USA. October 16, 2025.
UF URS 2025	Arushi Pant, Jessica Jiang, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Drosophila models of Amyotrophic lateral sclerosis (ALS)*, Undergraduate Research Symposium organized by University of Florida, Gainesville, Florida, USA. October 16, 2025.
IRCND 2025	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *A Drosophila-based genetic screen reveals new modifiers of Abeta and Tau co-pathology*, 5th International Research Conference on Neurodegenerative Diseases (IRCND 2025), Gainesville, Florida, USA. July 9–12, 2025.
IRCND 2025	Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, Yang S., Johnson A.N., and Diego E. Rincon-Limas, *Can SARS-Co-V2 Proteins Catalyze Abeta42-Induced Toxicity In Vivo?*, 5th International Research Conference on Neurodegenerative Diseases (IRCND 2025), Gainesville, Florida, USA. July 9–12, 2025.
AD/PD 2025	Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, Yang S., Johnson A.N., and Diego E. Rincon-Limas, *Can SARS-COV-2 proteins accelerate Abeta-42-induced neurodegeneration IN VIVO?*, 19th International Conference on Alzheimer's & Parkinson's Diseases Online, Vienna, Austria. April 1–5, 2025.
Neurology Day 2025	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *Discovering Novel Targets and Mechanisms Underlying Alzheimer Disease Pathogenesis in Drosophila*, Neurology Research Day held in Gainesville, Florida by Neurology Department, University of Florida. January 24, 2025.
Neurology Day 2025	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, Neurology Research Day held in Gainesville, Florida by Neurology Department, University of Florida. January 24, 2025.
ANA 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, 149th American Neurological Association conference held in Orlando, Florida, USA. September 14–17, 2024.
TAGC 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *A microRNA screening reveals miR-33 as a modifier of TDP-43 toxicity in flies*, TAGC-2024, held in Washington DC. March 6–10, 2024.
NCF SfN 2024	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *Discovering Novel Targets and Mechanisms Underlying Alzheimer's Disease Pathogenesis in Drosophila*, 14th Annual NCF SfN Regional Conference, University of Florida, FL, USA. February 25–26, 2024.
TAGC 2023	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates mutant TDP-43 toxicity in transgenic flies*, TAGC-2024, Chicago. March 1–5, 2023.

ICPMH 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behaviour in Drosophila*, International Conference on Psychology and Mental Health, Dubai, UAE. November 2021.
AAIC 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behaviour in Drosophila*, Alzheimer's Association International Conference Neuroscience Virtual. 2021.
IBRO-APRC 2021	**Shivam Kaushik** and Pinky Kain, *Understanding the role of taste circuits and its modulation using Drosophila melanogaster model*, Virtual IBRO-APRC Associate School of Neuroscience 2021 (**Best Oral Presentation Award**). 2021.
APDR5 2020	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, APDR5 Conference (IISER-PUNE). 2020.
SNCI 2019	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, SNCI 2019 Conference (Jamia Hamdard, New Delhi). 2019.
PAC 2019	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, PAC meeting of the Regional Centre for Biotechnology. 2019.

SKILLS

Drosophila Genetics

- **Genetic Workflows:** Handling of *Drosophila melanogaster* model, including complex crosses, mutant/transgenic line maintenance, and advanced systems (**GAL4-UAS, LexA/LexAop, RNAi, Optogenetics**).
- **Microdissections & Tissue Analysis:** Larval and adult dissections (*brain, gut, VNC, heart, wings, legs, and labellum*); phenotypic characterization and high-resolution imaging.
- **Feeding & Sensory Assays:** Proficient in **Two-choice, PER, CAFE** assays and spectrophotometric quantification for metabolic/feeding studies.
- **Behavioral Paradigms:** Expertise in locomotor performance and memory assays (**Fly-on-Ball, Zantiks** climbing/memory) and sensory response profiling.
- **Tissue Analysis: Calcium imaging, Two-Photon imaging, immunohistochemistry**, and quantitative analysis of neural and peripheral tissues.

Microscopy & Imaging

- **Advanced Microscopy: Two-photon** calcium imaging, **Confocal, Super-resolution** confocal, and **SEM** for neuronal activity and ultrastructural characterization.
- **Phenotypic Screening:** Fluorescence and stereo microscopy for tissue-level imaging, developmental staging, and phenotypic screening.
- **Acquisition:** Strong understanding of image acquisition parameters, optimization, and analytical workflows.

Molecular Biology Techniques

- **Pathway Analysis: PCR**, DNA isolation, gel electrophoresis, **Western blotting**, immunoblotting, and protein–protein interaction analysis.
- **Tissue Mapping: Immunohistochemistry (IHC)** of complex tissues (*brain, gut, heart, VNC*) with precise antibody-based localization and imaging.
- **Histology:** Tissue preparation including **fixation, dissection, sectioning, and staining** for high-resolution imaging.

Genetics (Classical & Molecular)

- **Core Genetics:** Genotype–phenotype correlation, tracking genetic lines, molecular genetics for functional gene analysis, and inheritance pattern analysis.

Bioinformatics Biology

- **Computational Analysis:** Sequence alignment (**BLAST, Clustal Omega, MUSCLE**), protein interaction prediction, and **AlphaFold2** structural modeling.
- **Data Integration:** Proficient in **NCBI, UniProt, FlyBase, and PDB**; skilled in integrating computational data with wet-lab findings.

Software & Data Analysis

- **Statistics & Languages:** MATLAB, GraphPad Prism, Origin, C++ (basic understanding), Biopython.
- **Scientific Illustration:** Adobe Illustrator, Photoshop, Premiere, ChemDraw, Microsoft Office Suite.

AWARDS & FELLOWSHIPS

- **Graduate Research Assistant**, *University of North Carolina, Chapel Hill, NC, USA*, 2022.
- **Presentation Award**, *Virtual IBRO-APRC Associate School of Neuroscience: Gut Microbiota and the Gut-Brain Connection*, 2021.

VOLUNTEER ACTIVITIES

Science Outreach & Leadership

- **Curriculum Development:** Designed and led the "We Are What We Eat" outreach program, focusing on nutrition and biological sciences for over 200 students. (Nov 2018)
- **Program Organizer:** Science outreach initiative focused on STEM education for secondary students at Government Sr. Sec. School, New Delhi. (July 2017)

Community Engagement

- **Outreach Member:** Organized and executed multi-site educational programs at Government Girls Inter College, Tehri, and Government Sr. Sec. School, Tehri, Uttarakhand, India. (Nov 22–23, 2018)
- **Educational Advocacy:** Promoted scientific literacy and health awareness in under-resourced regional schools through interactive workshops and seminars.